Masked Vision Transformer for Fast Hyperspectral Image Classification

Liguo Wang, *Member, IEEE*, Heng Wang, Shoulin Yin, and Lifeng Wang

***Abstract*— Vision Transformer (ViT) has been thoroughly explored in hyperspectral image (HSI) classification (HIC). Nevertheless, current ViT-based approaches still acquire discriminative features, resulting in relatively limited generalization capabilities when confronted with the challenges posed by high intraclass variances and interclass similarities commonly observed in HSI data. Moreover, most of these methods fail to adequately emphasize the significance of the central pixel in HIC. To address the aforementioned challenges, we introduce a masked ViT (MViT) for HIC. First and foremost, MViT endeavors for the first time to introduce the masking operations in supervised models to learn more robust patterned features instead of distinguishable features, thereby bestowing it with outstanding generalization performance. Second, during the training phase of MViT, when conducting the random masking operations on the embedded features, we deliberately retain the embedding corresponding to the central pixel to guarantee the effectiveness of the model and emphasize the importance of the central pixel in HIC. Finally, MViT will deactivate the masking operations during the testing phase and utilize all the embedded features to accomplish the classification task, thereby enabling the model to fully exploit its recognition capabilities. On top of that, MViT is an extremely lightweight model, and by introducing the masking operations during the training phase, its training speed becomes unprecedentedly rapid. Experiments conducted on four publicly accessible datasets demonstrate that MViT can consistently achieve excellent or even the optimal classification results in comparison with the most advanced methods. The source code was powered by Jupyter and released at https://github.com/swiftest/MViT.**

***Index Terms*— Hyperspectral image (HSI) classification (HIC), lightweight model, masked Vision Transformer (MViT), masking operations, patterned features.**

I. Introduction

HYPERSPECTRAL images (HSIs) naturally possess rich spectral and spatial information. As a fundamental task in HSI data analysis, HSI classification (HIC) plays an irreplaceable role in agriculture monitoring, mineral exploration, environmental supervision, and so on [1], [2], [3], [4]. The rapid development of deep learning has provided a continuous stream of new technological support for HIC, and many excellent deep models have been developed [5], [6], [7]. It is worth noting that in the presence of data-hungry deep models, the issue of the dimensional curse resulting from spectral redundancy is no longer a concern that needs to be emphasized. Data dimensionality reduction is frequently employed to guarantee efficient training and testing of deep models, and some models [8], [9], [10], [11] inherently possess a large model capacity, enabling them to accommodate the full spectral information.

Convolutional neural networks (CNNs) shine in the field of HIC. Roy et al. [5] established a hybrid spectral CNN (HybridSN) to investigate the stacked 3D–2D CNN hierarchy for HIC. Qiao et al. [12] designed a lightweight network based on spectral and channel attention modules, namely LiteSCANet. LiteSCANet utilizes a residual double-branch unit to acquire spectral–spatial features, and employs channel and spectral attention modules to study the interdependencies among different channels and the interactions among spectral bands. Considering that the self-attention mechanism-based Transformer models merely possess spatial adaptability while disregarding channel adaptability, Wu et al. [13] proposed a novel spectral–spatial large kernel attention (SSLKA) network. SSLKA employs two synergistic spectral–spatial attention blocks featuring large convolution kernels to effectively extract spectral–spatial joint features. Roy et al. [14] used 3-D convolution kernels with different receptive fields to mine adaptive spectral–spatial joint features, and combined residual blocks with different structures to construct the attention-based adaptive kernel residual network (A2S2K-RN). Meng et al. [15] devised a lightweight spectral–spatial convolution module (LS2CM) for HIC, which is capable of maintaining or even enhancing the classification performance while significantly reducing network parameters and floating-point operations (FLOPs). As a plug-and-play component, LS2CM can be effortlessly utilized to upgrade existing CNN-based models. Chen et al. [16] constructed a new two-branch embedded multivariate attention network (DMCN) for HIC. In DMCN, the spectral branch is dedicated to extracting spectral features by employing multiscale convolution and a normalization-based attention module, while the spatial branch captures spatial features via small-scale dilation convolution and Euclidean similarity attention. Ye et al. [17] designed a multiscale channel attention and multiscale depthwise

Received 8 April 2025; revised 5 May 2025; accepted 18 May 2025. Date of publication 21 May 2025; date of current version 29 May 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62071084 and Grant 62301054 and in part by the Fundamental Research Funds for the Central Universities under Grant 04442024040 and Grant 04442024041. *(Corresponding author: Heng Wang.)*

Liguo Wang is with the College of Information and Communication Engineering, Dalian Nationalities University, Dalian 116600, China (e-mail: wangliguo@hrbeu.edu.cn).

Heng Wang and Shoulin Yin are with the College of Information and Communication Engineering, Harbin Engineering University, Harbin 150001, China (e-mail: wanghenghrbeu@hrbeu.edu.cn; yslin@hit.edu.cn).

Lifeng Wang is with the School of Technology, Beijing Forestry University, Beijing 100083, China (e-mail: wanglifeng_2016@163.com).

Digital Object Identifier 10.1109/TGRS.2025.3572242

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:20:51 UTC from IEEE Xplore. Restrictions apply.

convolution network, namely MSCA-MSDC. This network deploys multiscale channel attention to capture more abundant channel information, and extracts multiscale spatial features through the utilization of multibranch depthwise convolution (DW-Conv) in the form of channel partitioning. To optimize the widely used DW-Conv and pointwise convolution (PW-Conv) in lightweight models, Ma et al. [18] proposed an ultra-lightweight hybrid CNN based on redundancy removal (ULite-R2HCN). ULite-R2HCN utilizes a specially engineered spatial convolution that incorporates redundancy elimination to minimize redundant convolution kernels and extract multiscale spatial features, and employs a spectral convolution with redundancy removal to circumvent unnecessary spectral extractions while capturing long-range spectral relationships. Despite its lightweight design, ULite-R2HCN's performance is not particularly remarkable. Considering the substantial number of repetitive computations in patch-based classifiers, Zheng et al. [19] put forward a fast patch-free global learning framework (FPGA). FPGA can fully exploit the global spatial information, and the spectral attention module within this framework effectively models the interdependencies of the feature maps. Tang et al. [8] proposed a spatial–spectral attention-in-attention network (S2AiANet), which incorporates a spatial–spectral attention-in-attention (S2AiA) module into the global learning framework to concurrently take into account spatial and spectral information. Meanwhile, the network introduces a multiscale attention (MSA) module to clearly distinguish the weights of the features at different scales. Both S2AiANet [8] and FPGA [19] belong to global learning frameworks, but the problem is that in practical applications, the number of labeled samples we can access is limited after all, and a global learning framework will carry out unnecessary calculations at a large number of locations during the training process.

Although the pooling operations in CNN can effectively detect the appearance of certain features, the relative spatial position information of these features has not been fully taken into account. Paoletti et al. [20] initially introduced the capsule network (CapsNet) into HIC to achieve more accurate classification. Considering the lack of global attention mechanism in CapsNet, Wang and Wang [21] introduced Vision Transformer (ViT) [22] into CapsNet and built CapsViT classifier. Ravikumar et al. [23] introduced the second-generation CapsNet, the deep matrix capsule network (DMCN) [24], to HIC. DMCN is a highly suggestive model that initially computes the votes of convolutional capsules by multiplying the primary capsule with the viewpoint-invariant transformation matrix, followed by determining the Gaussian mixture distributions of these convolutional capsules through assignment probabilities and votes. Ultimately, based on the similarity between votes and distributions, primary capsules are aggregated into convolutional capsules to effectively model part-whole relationships. While both CapsViT [21] and DMCN [23] demonstrate commendable classification performance, they incorporate intricate routing mechanisms that contribute to increased model complexity.

Graph convolution networks (GCNs) have the ability to model the long-range relations among samples in HSIs, which is not achievable in CNNs. Hong et al. [25] designed a novel mini-batch GCN (miniGCN), integrating features extracted from the CNN stream and the GCN stream to achieve better classification results. Wan et al. [26] introduced a multiscale dynamic GCN to benefit both graph convolution and graph update, and gradually generate discriminable embedded features and refined graphs. Additionally, to comprehensively utilize the multiscale spectral–spatial information, the authors also explore multi-input graphs with different neighborhood scales. Jiang et al. [27] proposed a graph generative structure-aware Transformer (GraphGST) that employs a positional encoder and stacked Transformer encoders to offer local and global views on pixel representation learning, and uses a cross-view contrast mechanism to enable the local and global views to supervise and complement each other. Zhao et al. [28] proposed the spectral–spatial graph attention network (SSGAT) that generates more discriminable aggregate features through calculating the attention coefficients between a node and all nodes within a large neighborhood. While GCNs appear to hold significant promise, the process of graph construction does not inherently prioritize the impact of uncertainty on classification.

With the ascent of the Transformer architecture [29] in the domain of natural language processing (NLP), Dosovitskiy et al. [22] correspondingly put forward ViT in the field of computer vision, and demonstrated that a pure Transformer applied directly to sequences of image patches could perform extremely well on image classification tasks. For HIC tasks, the combination of ViT and CNNs is becoming predominant [30], [31], [32], [33]. Inspired by the potent learning capacity of ViT for global dependencies, He et al. [34] investigated a new framework called transferring spatial–spectral Transformer (T-SST). What is creative is that T-SST employs a VGG-like deep CNN to extract spatial features and utilizes densely linked ViT blocks to learn long-range dependencies among channels. To comprehensively mine and represent the sequence properties of the sample cubes, Hong et al. [35] designed SpectralFormer, a model takes the groupwise spectral embeddings as the input tokens of ViT encoders and fuses the outputs of encoders at different depths through a cross-layer adaptive fusion (CAF) strategy to enhance information exchange. Although SpectralFormer has some novelty, as a spectral sequence processing model, its performance is not satisfactory. Zhang et al. [36] proposed an effective lightweight spectral–spatial Transformer (ELS2T) to address the issue of low computational efficiency and the incapability of effectively fusing spectral–spatial information in existing CNN-based models. ELS2T uses a global MSA module and an adaptive feature fusion module to acquire spectral–space joint features, and utilizes the elaborately designed lightweight separable spatial–spectral self-attention (S3A) module to replace the multihead self-attention (MHSA) module to improve the computational efficiency. Sun et al. [37] presented a spectral–spatial feature tokenization Transformer (SSFTT) to emphasize the learning of high-level semantic features. What sets SSFTT apart is that it uses a Gaussian-weighted feature tokenizer to transform the shallow spectral–spatial features extracted by the convolutional module into high-level semantic

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:20:51 UTC from IEEE Xplore. Restrictions apply.

tokens, which are then fed into ViT to accomplish the classification task. To its credit, SSFTT is not only lightweight, but also demonstrates exceptional performance. Mei et al. [38] developed a group-aware hierarchical Transformer (GAHT) to investigate the application of hierarchical Transformer in HIC. Ouyang et al. [39] developed HybridFormer by integrating multigranularity semantic token embedding with spatial–spectral feature modeling. A key innovation of this model lies in its use of multiscale depthwise separable convolution to learn multifine-grained semantic token embeddings, which serve as inputs for the ViT module. Chen et al. [40] reevaluated the role of the center pixel in ViT-based models and devised a spectral query spatial Transformer (SQSFormer) framework. SQSFormer combines random center rotation and center relative position embedding to mitigate the interference of absolute position information on spatial features extraction and employs multiple stacked center pixel embeddings instead of the class token to complete the classification task. To improve the complexity of the existing ViT-based models, Gu et al. [41] put forward a multiscale spectral–spatial attention network with frequency-domain lightweight Transformer (MSA-LWFormer). MSA-LWFormer utilizes the fast Fourier transform (FFT) to convert the spatial-domain features into the frequency-domain counterparts during the self-attention calculation stage, which facilitates the integration of global information and significantly lowers the time complexity of the network. To alleviate the potential depth decay, Sun et al. [42] built a memory-augmented spectral–spatial Transformer (MASSFormer). MASSFormer retains memory by concatenating the memory tokens acquired through pooling operations to the end of the keys and values in each encoder's MHSA module, which allows the model to achieve adequate information mixing. Wu et al. [43] put forward a dilated deep multiscale pooling-based MetaFormer network (DDMN) to strengthen the exploitation of the spectral information. DDMN relies on the dilated depthwise separable convolution to capture more detailed spectral features, and employs the multiscale pooling-based MetaFormer [44] instead of ViT to reduce the number of parameters and accelerate training. Zhao et al. [45] proposed a novel dual-branch spectral–spatial attention network (DBSSAN) to stress the significance of the central pixel. DBSSAN feeds the central pixel into ViT to learn long-distance dependencies and nonlinear correlations in HSI data, and uses a spatial attention module based on cosine similarity and Gauss–Euclidean similarity, along with a scale information extraction (SIE) module to capture local and global spatial features. Although DBSSAN has been proved to be efficient, from the aspect of model complexity, DBSSAN demands considerable training and testing time since it is computationally intensive to acquire the spatial attention maps.

Although the current ViT-based models have achieved significant progress, they still learn discriminative features, which leads to their relatively poor generalization ability when facing the challenges brought by the high intraclass differences and interclass similarities commonly found in HSI data. Additionally, the majority of the current approaches have not fully considered the importance of the central pixel in HIC. To tackle the aforementioned issues and further improve the performance of HIC, we propose a lightweight masked ViT (MViT) that leverages the perspective of acquiring patterned features. It is essential to emphasize that the so-called patterned features refer to the intrinsic patterns or universal representations that are independent of specific input samples. Frankly speaking, MViT no longer prioritizes the extraction or identification of the discriminative features used for distinguishing samples across various categories. Instead, it emphasizes the learning of the internal patterns. This characteristic enables MViT to exhibit superior robustness when facing the HIC challenges posed by the high intraclass variances and interclass similarities.

For the proposed MViT, during the training phase, it first employs a 3-D convolution layer and a 2-D convolution layer to extract spectral–spatial features, and then expands the obtained features in the spatial dimension. Next, MViT utilizes a random masking operation with a masking ratio of 75% to discard a large number of features other than the embedded feature corresponding to the central pixel. This strategy not only emphasizes the importance of central pixel in HIC, but also ensures the effectiveness of the model. Finally, all the remaining embedded features and an additional class token are fed into the ViT blocks to learn the optimal representation for the final classification task. We assert that MViT acquires more robust patterned features via the masking operations rather than traditional discriminative features, thus enabling it to possess better generalization performance. During the testing phase, MViT will turn off the masking operations and utilize all the embedded features to fulfill the classification task, giving full play to its recognition capabilities. MViT is a very lightweight model, and its advantages lie not only in its ability to learn more robust patterned features, but also in its astonishingly fast training speed due to the introduction of the mask operations during the training phase.

The key contributions of this work are as follows.

1) We have designed a lightweight MViT for HIC. For the first time, MViT incorporates the masking operations in supervised classification models to learn more robust patterned features rather than discriminative features in the traditional sense, which endows it with better generalization performance.
2) MViT adopts different feature processing modes during the training and testing phases. During the training phase, to emphasize the contribution of the central pixel in HIC and ensure the effectiveness of the entire model, MViT will deliberately retain the embedded feature corresponding to the central pixel when performing the mask operations. During the testing phase, to maximize the recognition capabilities of the model, MViT will deactivate the masking operations and use all the embedded features to complete the classification task.
3) MViT is not only a lightweight model, but also has a very fast training speed due to the introduction of the masking operations during the training phase.

The rest of this article is organized as follows. Section II describes the overall framework of MViT in detail, and presents the design concept. The experimental results and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:20:51 UTC from IEEE Xplore. Restrictions apply.

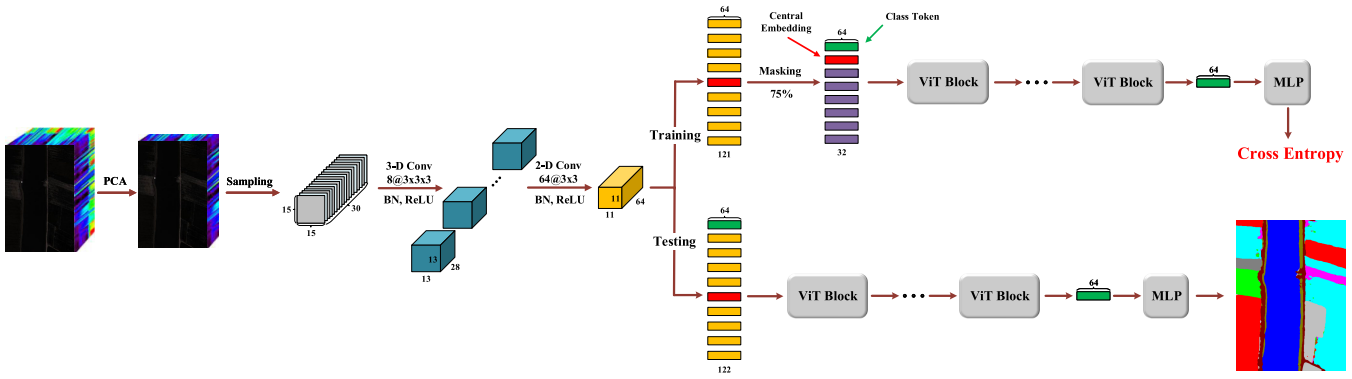


Fig. 1. Overview of the proposed MViT, where PCA refers to the principal component analysis, BN indicates the batch normalization, ReLU represents the rectified linear unit, and MLP stands for the multilayer perceptron.

analysis are presented in Section III. Section IV discusses the complexity and sensitivity of MViT. Finally, Section V summarizes this article and gives the possible research direction.

II. Methodology

In this section, we initially introduce how MViT operates, and subsequently provide corresponding design concepts to facilitate understanding of its effectiveness.

A. Overall Framework

For ease of understanding, we present the overview of MViT in Fig. 1. As shown in Fig. 1, MViT adopts different feature processing modes during the training and testing phases. Let us start with the training phase of MViT.

Let $\boldsymbol{X} \in \mathcal{R}^{B\times H\times W}$ represent the original HSI data with height H, width W, and B spectral channels. First, PCA (principal component analysis) is applied to effectively reduce the data dimensionality. After dimensionality reduction, 30 bands are retained. We use $\boldsymbol{X}' \in \mathcal{R}^{30\times H\times W}$ to represent the data at this time. Subsequently, we select the samples with a spatial size of 15×15 centered on the target pixel as the input of MViT. The reason for setting the spatial size of the samples to 15×15 lies in the fact that a large spatial size would result in a considerable amount of redundant calculations, while a small spatial size would restrict the spatial information encompassed within the samples. Let $\boldsymbol{x} \in \mathcal{R}^{N\times 1\times 30\times 15\times 15}$ represent the input HSI samples in one batch. MViT first uses a 3-D convolution layer to extract spectral–spatial joint features

$$\boldsymbol{e} = \text{ReLU}(\text{Conv3d}(1, 8, 3)(\boldsymbol{x})) \tag{1}$$

where ReLU represents the rectified linear unit activation function, and $\boldsymbol{e} \in \mathcal{R}^{N\times 8\times 28\times 13\times 13}$ represents the embedded features obtained. Then, the obtained features are cascaded in the channel dimension and fed into a 2-D convolutional layer to continue learning advanced embedded features

$$\boldsymbol{e}' = \text{ReLU}(\text{Conv2d}(224, 64, 3)(\text{Cascade}(\boldsymbol{e}))) \tag{2}$$

where $\text{Cascade}(\boldsymbol{e}) \in \mathcal{R}^{N\times 224\times 13\times 13}$ denotes the cascading operation on $\boldsymbol{e}$, and $\boldsymbol{e}' \in \mathcal{R}^{N\times 64\times 11\times 11}$ stands for the output of the 2-D convolution layer. Next, $\boldsymbol{e}'$ is unfolded in the spatial dimension, i.e., $\boldsymbol{e}' \in \mathcal{R}^{N\times 121\times 64}$. To represent the relative spatial positions of all the embedded features, $\boldsymbol{e}'$ is added with a fixed position embedding $\mathbf{p} \in \mathcal{R}^{1\times 121\times 64}$ calculated by the cosine function used in the Transformer framework [29]

$$\boldsymbol{e}'' = \boldsymbol{e}' + \mathbf{p}. \tag{3}$$

To emphasize the contribution of the central pixel in HIC and ensure the effectiveness of the entire model, MViT will deliberately retain the embedding corresponding to the central pixel before performing the masking operation. Here, we refer to the embedding corresponding to the central pixel as the central embedding and represent it as $\boldsymbol{c}$

$$\boldsymbol{c} = \boldsymbol{e}''[:, 60, :].\text{unsqueeze}(1) \tag{4}$$

and we represent the embedding features in $\boldsymbol{e}''$ other than the central embedding as $\boldsymbol{d}$

$$\boldsymbol{d} = \text{Cat}\big(\big[\boldsymbol{e}''[:, :60, :], \boldsymbol{e}''[:, 60:, :]\big], \text{dim} = 1\big) \tag{5}$$

where "unsqueeze" indicates the dimension expansion operation, and "Cat" represents the concatenation operation. At this point, $\boldsymbol{c} \in \mathcal{R}^{N\times 1\times 64}$ and $\boldsymbol{d} \in \mathcal{R}^{N\times 120\times 64}$. The central embedding $\boldsymbol{c}$ is always preserved, and MViT only performs the random masking operation on $\boldsymbol{d}$ with a default masking ratio of 75%

$$\boldsymbol{d}' = \text{Masking}(\boldsymbol{d}) \tag{6}$$

where "Masking" stands for the random masking operation, and $\boldsymbol{d}' \in \mathcal{R}^{N\times 30\times 64}$. Note that the 75% masking ratio is merely a default value and not necessarily the optimal one. We will discuss the sensitivity of MViT to different masking ratios in Section IV.

Let $\boldsymbol{cls_token} \in \mathcal{R}^{1\times 1\times 64}$ represent the class token (its position embedding is $\mathbf{0}^{1\times 1\times 64}$), then the input of the first ViT block can be expressed as follows:

$$\boldsymbol{f} = \text{Cat}\big(\big[\boldsymbol{cls_token}.\text{repeat}(N, 1, 1), \boldsymbol{c}, \boldsymbol{d}'\big], \text{dim} = 1\big) \tag{7}$$

where "repeat" represents an operation that copies the tensor in a particular dimension, and $\boldsymbol{f} \in \mathcal{R}^{N\times 32\times 64}$. In the default configuration of MViT, there are 3 ViT blocks, and the class token output from the last ViT block is fed into the MLP (multilayer perceptron) module to complete the training task. It is important to highlight that MViT establishes the number

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:20:51 UTC from IEEE Xplore. Restrictions apply.

of ViT blocks at 3, both to ensure the model's depth and to manage its complexity.

During the testing phase, MViT will turn off the masking operations and use all the embedded features to complete the classification task. Specifically, during the testing phase, the input of the first ViT block will be

$$\boldsymbol{f} = \mathrm{Cat}\left(\left[\boldsymbol{cls_token}.\mathrm{repeat}(N, 1, 1), \boldsymbol{e}''\right], \mathrm{dim} = 1\right) \quad (8)$$

where $\boldsymbol{f} \in \mathcal{R}^{N\times 122\times 64}$. Using all the embedded features for prediction will fully exert the recognition capabilities of MViT.

B. Design Concept

We say that MViT learns patterned features rather than discriminable features in the traditional sense. In other words, in HIC, pattern features are more robust than discriminable features, which gives MViT better generalization capabilities. Here, we illustrate the effectiveness and rationality of MViT from the perspectives of masked image modeling (MIM) and Dropout.

1) MIM Perspective: Inspired by BERT (bidirectional encoder representations from Transformer) [46], a masked language representation model, He et al. [47] proposed masked autoencoder (MAE) by analyzing the differences between visual and language masked autoencoding. The great success of MAE lies in the fact that it creates an extremely challenging self-supervisory task, namely, reconstructing a large number of masked patches by using a small number of visible patches' tokens along with the latent representation. In our previous research [48], through a profound analysis of the principles underlying MAE, we developed a self-supervised representation learning model known as masked CNN (MCNN), and demonstrated that masking operations (a specialized form of pooling) contribute to the discovery of patterns. We claim that the pooling operations function as a local attention mechanism, Transformer serves as a global attention mechanism, and the masking operations are indispensable for learning patterned features. MViT utilizes the masking operations with a high masking ratio to randomly discard a substantial number of embedded features, effectively facilitating the learning of patterned features for classification, and the reason why MViT sets the default masking ratio to 75% is also based on the research findings of MAE. It is noteworthy that MViT disables the masking operations during the testing phase and utilizes all embedded features to accomplish the classification task. This strategy draws inspiration from MAE, which employs all patches to learn the representations of samples in the fine-tuning stage. By doing so, MViT aims to avoid the introduction of uncertainties in predictions while fully leveraging all embedded features to maximize its recognition capabilities.

From the perspective of the masking mechanism, MViT learns the patterned features by retaining only the central embedding while discarding a significant number of spectral–spatial embeddings. In contrast, MAE learns the representations by masking a large proportion of patch embeddings. Both of them aim to uncover the intrinsic patterns of samples in a manner that transcends differences and identifies similarities. However, a key distinction lies in their learning paradigms: MViT is a supervised classification model, whereas MAE is a self-supervised (unsupervised) representation learning model.

Why cannot the central embedding be discarded? First, if the central embedding is discarded randomly during training, although MViT will still converge, its training process may become oscillatory, meaning it will not converge smoothly (see Section IV-F for a detailed stability analysis). Second, regarding HIC, the samples we use are sample cubes centered around the target pixel, and the central pixel indeed plays a decisive role in the final classification. If the central embedding is abandoned, the class token learned by MViT will be solely based on neighborhood information. Nevertheless, samples from different categories might have similar neighborhood information, and MViT is prone to misclassify these samples.

2) Dropout Perspective: MViT employs the masking operations to randomly discard features and will disable the masking operations during the testing phase. These features bring to mind Dropout. 1-D Dropout randomly discards a portion of the elements in the feature vector, while 2-D Dropout randomly discards a subset of elements within the feature map across each channel. By analogy, the random masking operation employed by MViT involves the stochastic removal of certain spectral–spatial embeddings. Dropout has been proven effective in neural networks and can significantly enhance the generalization performance of a classification model. Thus, MViT also enjoys such advantages. We say that Dropout is actually attempting to learn patterned features. However, it should be noted that MViT does not divide the retained embedded features by the retention probability after the masking operation. This is because MViT attaches greater importance to the contribution of the central pixel. As long as the central embedding is not discarded, the aforementioned operation is not necessary.

III. Experimental Results and Analysis

We use the WHU-Hi dataset and the Pavia University (PU) dataset to evaluate the effectiveness and superiority of MViT. Among them, the WHU-Hi dataset was obtained using a headwall nano-hyperspec sensor in the farming areas featuring various crop types in Hubei Province, China. This dataset encompasses three scenes, namely LongKou, HanChuan, and HongHu, and possesses the following characteristics: 1) extremely high spatial resolution; 2) an extremely high proportion of labeled pixels; and 3) abundant ground types. The WHU-Hi dataset is a challenging one, while the PU dataset is one of the most classic ones in the remote sensing community. Experiments conducted on these datasets can more effectively evaluate the performance of MViT, and the compared models include A2S2K-RN [14], CapsViT [21], SpectralFormer [35], SSFTT [37], GAHT [38], HybridFormer [39], SQSFormer [40], and MASSFormer [42]. Typically, we use the overall accuracy (OA), the average accuracy (AA), and the kappa coefficient (κ) to assess the classification performance. All the experiments were carried out on a system equipped with an NVIDIA GeForce RTX 3090 GPU accelerator (with 24-GB GPU memory).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:20:51 UTC from IEEE Xplore. Restrictions apply.

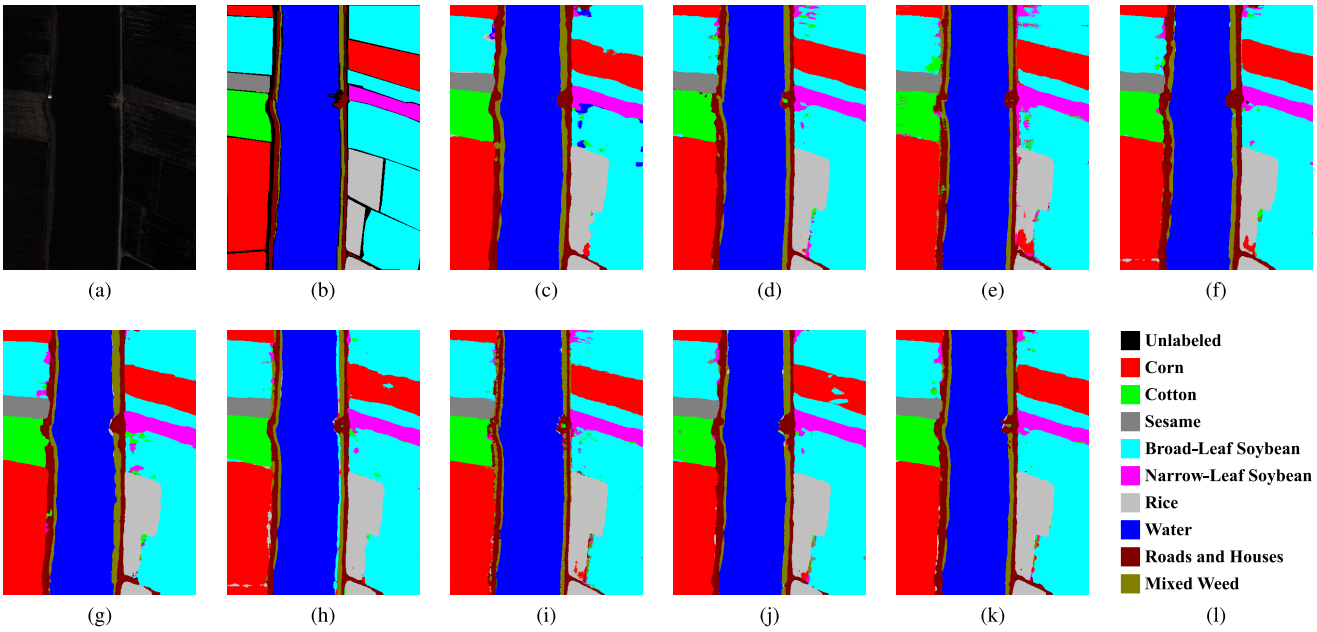


Fig. 2. Visualization of the classification maps obtained by different models on the LongKou dataset. (a) Three-band false color composite. (b) Ground truth. (c)–(k) Classification maps obtained by A2S2K-RN, CapsViT, SpectralFormer, SSFTT, GAHT, HybridFormer, SQSFormer, MASSFormer, and MViT. (l) Legend.

A. Experimental Setup

First, the spatial sizes of the samples utilized in all models were set at 15×15, as an excessively large spatial size results in a significant amount of redundant calculations, whereas a spatial size that is too small restricts the amount of spatial information contained within the samples. Second, PCA was employed to reduce the dimensionality of all the datasets to 30 bands. The dimensionality reduction via PCA is indispensable since it enables various models to avoid the potential problem of the dimensional curse and significantly boosts their training and testing speed. It is worth highlighting that the aforesaid settings might not be the most optimal for some models, thus, we need to make a few minor adjustments to those models. For models that require minor adjustments, the primary adjustments involve modifying the input channels and convolutional strides of the preceding convolutional layers in accordance with the spatial size and channel of the input sample, while the configurations of the other model components (layers or modules) remain unaltered.[1] Finally, for each dataset, we randomly picked 25 (labeled) samples from each category as training samples, and the remaining ones were used as test samples. Additionally, to obtain the optimal parameters of each model, we randomly selected 100 samples from each category within the test set to construct the validation set. All models were trained for 100 epochs with a batch size of 32. To prevent the number of samples in the last mini-batch from being 1, we sometimes adjusted the batch size to 30. For our approach, the Adam optimizer was employed to update the network parameters, with an initial learning rate of 0.001, and an exponential learning rate schedule with an attenuation factor of 0.99 is employed to gradually lower the learning rate as the training proceeds.

Since the selection of labeled samples will undoubtedly have a significant impact on the performance of the model, to comprehensively assess the performance of each model on each dataset, we repeated the experiment ten times for each model to report the AA and standard deviation.

[1]The detailed numerical adjustments will not be exhaustively listed here, and interested readers may refer to our open source code.

B. Experiment 1: LongKou Dataset

The scene of LongKou was acquired in 2018, it is a simple agricultural region and contains 9 land-cover species. The imagery contains 550×400 pixels, there are 270 bands from 400 to 1000 nm, and the spatial resolution is about 0.463 m. Fig. 2(a) exhibits the false-color composite image, and Fig. 2(b) shows the ground-truth map.

We present the classification results of diverse models on the LongKou dataset in Table I. Obviously, SQSFormer had achieved the best results on the LongKou dataset. This indicates that the random center rotation and center relative position embedding employed in SQSFormer can effectively mitigate the interference of absolute position information on spatial feature extraction and render the model more robust. By looking at the data in Table I, it can be observed that class 8 (Roads and Houses, brown) is the most challenging to identify for all the models, and SQSFormer has the highest recall rate for class 8 (95.12%), which is the primary reason for its outstanding performance on the LongKou dataset. A2S2K-RN, which extracts multiscale adaptive features and introduces the attention mechanism, achieved satisfactory classification results. Among all ViT-based models, SpectralFormer performs relatively poorly as it is merely a simple spectral sequence processing model. Since CapsViT introduces ViT into CapsNet to model the global correlations among all

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:20:51 UTC from IEEE Xplore. Restrictions apply.

TABLE I
CLASSIFICATION RESULTS OF VARIOUS MODELS ON THE LONGKOU DATASET

Class		A2S2K-RN	CapsViT	SpectralFormer	SSFTT	GAHT	HybridFormer	SQSFormer	MASSFormer	MViT
Accuracies(%)	C1	**99.74±0.24**	99.67±0.22	97.55±2.14	99.20±0.51	98.93±0.82	99.30±0.75	99.42±0.46	99.65±0.30	99.52±0.40
	C2	98.11±1.22	98.20±1.21	94.81±1.62	97.84±1.06	97.67±1.60	97.04±1.64	98.01±0.92	97.36±1.96	**99.05±0.70**
	C3	99.75±0.41	99.79±0.27	99.33±0.44	99.51±0.43	99.77±0.44	**99.87±0.21**	99.84±0.34	99.83±0.21	99.56±0.38
	C4	**97.02±1.53**	95.02±1.99	93.01±1.64	95.27±1.67	95.86±1.30	95.43±1.15	96.00±1.77	96.54±1.07	96.32±1.66
	C5	99.36±0.67	**99.52±0.54**	98.37±1.81	98.39±1.24	99.16±0.87	99.27±0.86	99.27±0.62	99.36±0.72	99.27±0.82
	C6	97.90±1.34	99.22±1.00	98.53±1.24	98.32±1.31	97.76±1.15	99.01±0.72	**99.65±0.31**	98.00±2.43	99.38±0.57
	C7	96.35±1.64	98.79±0.60	98.58±0.70	97.73±1.02	97.35±1.28	98.60±1.26	**99.55±0.57**	98.79±0.55	98.94±0.64
	C8	85.76±5.40	94.01±2.04	88.20±3.08	90.34±3.85	89.47±5.48	94.48±2.48	**95.12±0.44**	87.67±3.82	92.73±2.61
	C9	87.98±5.29	95.97±1.84	**96.21±2.84**	88.45±3.95	92.80±4.14	95.86±2.38	95.58±1.61	92.86±2.60	93.34±6.26
OA(%)		96.82±0.74	97.57±0.62	96.11±0.85	96.80±0.65	96.87±0.59	97.52±0.55	**98.12±0.53**	97.63±0.36	97.91±0.63
AA(%)		95.78±0.54	97.80±0.34	96.07±0.38	96.12±0.65	96.53±0.44	97.65±0.45	**98.05±0.30**	96.67±0.44	97.57±0.68
κ		0.9586±0.009	0.9682±0.008	0.9494±0.011	0.9583±0.008	0.9592±0.008	0.9676±0.007	**0.9754±0.007**	0.9689±0.005	0.9727±0.008

primary capsules, it can be seen that it has an extremely well performance. As a lightweight model, SSFTT can achieve an OA of 96.80%, an AA of 96.12%, and a κ of 0.9583, which is highly praiseworthy. To highlight the problem of information loss after cascading multiple ViT blocks, MASSFormer integrates memory tokens into the MHSA module of each encoder. Based on the experimental results, it attained the third-best performance, following SQSFormer and MViT. GAHT is a layered Transformer, at each layer, it employs the grouped pixel embedding (GPE) module and the Transformer encoders to acquire local–global spectral–spatial features. Although the notion underlying GAHT is novel, its performance is not remarkable and is quite close to that of SSFTT and A2S2K-RN. HybridFormer employs a multigranularity token generator (MTG) to learn multigranularity spectral–spatial embedding tokens. Subsequently, it leverages a novel transformer encoder to capture global and local information of input tokens across different granularity branches. Finally, the outputs from these granularity branches are adaptively weighted and fused to produce the final classification result. Based on the classification results, HybridFormer demonstrates performance comparable to CapsViT and is equally outstanding. Most notably and most thrillingly, MViT achieved a performance second only to SQSFormer. We assert that MViT acquires the input-independent patterned features through the masking operations rather than the distinguishable features in the traditional sense, which enables it to have outstanding generalization performance.

After quantitatively evaluating each model, let us use the classification diagrams to qualitatively assess their performance. We set a random seed (42) to select fixed training samples and trained each model. The obtained classification maps are shown in Fig. 2. Visually speaking, SQSFormer remains one of the finest. In the classification map obtained by SpectralFormer, a significant number of samples from class 6 (Rice, light gray) are misclassified as belonging to class 1 (Corn, red). It is notable that CapsViT had obtained a highly distinctive classification graph, which once again validates its ability to achieve high classification accuracy on the LongKou dataset. By observing the classification diagram obtained through GAHT, we can discover that if it does not misclassify numerous samples belonging to category 4 (Broad-Leaf Soybean, light blue) as category 5 (Narrow-Leaf Soybean, pink), the visual effect of the classification diagram becomes extremely well. To its credit, the classification map obtained by HybridFormer is clean and well-defined. Finally, we can see that the classification map obtained by the proposed MViT contains relatively few noise points, precisely verifying its effectiveness and superiority.

C. Experiment 2: HanChuan Dataset

The scene of HanChuan was acquired in 2016 and the study area is planted with 16 ground types. The size of the imagery is 1217 × 303, there are 274 bands from 400 to 1000 nm, and the spatial resolution is about 0.109 m. Notably, there are many shadow-covered areas in the imagery, which brings great challenges to the classification task. Fig. 3(a) and (b) shows the false-color composite image and the ground truth.

Accordingly, the classification results obtained by each model on the HanChuan dataset are given in Table II. As indicated in Table II, A2S2K-RN obtained the best classification results on the HanChuan dataset. The proposed MViT, although not as impressive as A2S2K-RN, achieved the second-best performance, with an average OA 0.53% lower than that of A2S2K-RN. It is notable that although SQSFormer obtained the best classification results on the LongKou dataset, its performance on the HanChuan dataset was rather mediocre. Likewise, CapsViT and HybridFormer are no longer as prominent as they appear on the LongKou dataset. SQSFormer and MViT stress the significance of the central pixel in HIC in diverse manners: SQSFormer employs random rotation and relative position embedding to accentuate the importance of the central pixel. Significantly, SQSFormer stacks the output of each Transformer encoder for the central pixel embedding to fulfill the classification task instead of aggregating a class token; MViT underlines the importance of the central pixel by preserving the central embedding when conducting the random masking operations. Essentially, SQSFormer and MViT employ Transformer for distinct reasons. SQSFormer discards Transformer's aggregate learning capabilities (learning a class token) and merely utilizes it as a global attention mechanism, while MViT exploits Transformer's aggregate learning capabilities to acquire patterned features for classification. From the experimental results, it is evident that MViT is significantly superior to SQSFormer, indicating that MViT can achieve consistently outstanding performance when dealing with different datasets. SpectralFormer continues to be the model

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:20:51 UTC from IEEE Xplore. Restrictions apply.

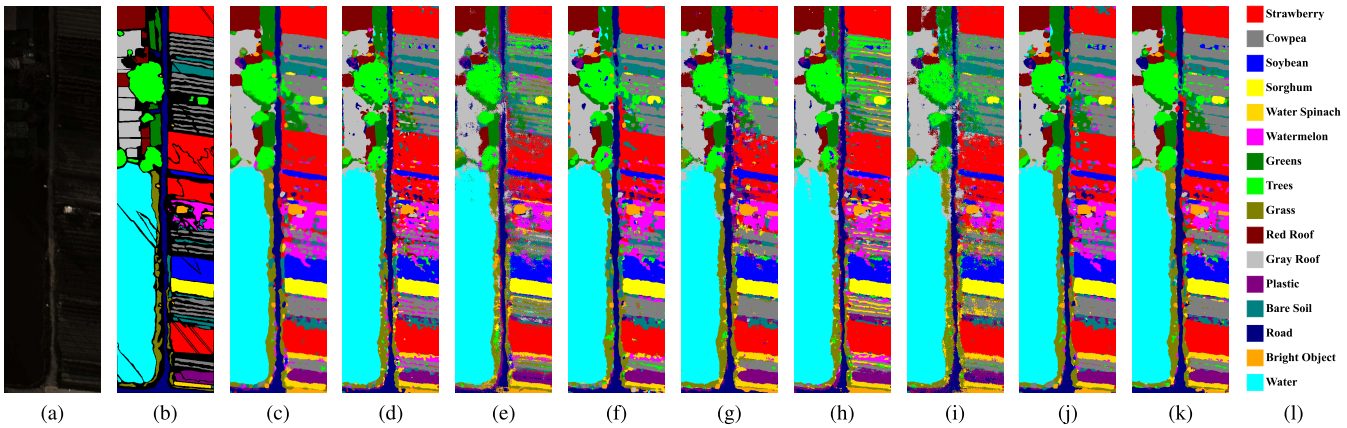


Fig. 3. Visualization of the classification maps obtained by different models on the HanChuan dataset. (a) Three-band false color composite. (b) Ground truth. (c)–(k) Classification maps obtained by A2S2K-RN, CapsViT, SpectralFormer, SSFTT, GAHT, HybridFormer, SQSFormer, MASSFormer, and MViT. (l) Legend.

TABLE II
CLASSIFICATION RESULTS OF VARIOUS MODELS ON THE HANCHUAN DATASET

Class		A2S2K-RN	CapsViT	SpectralFormer	SSFTT	GAHT	HybridFormer	SQSFormer	MASSFormer	MViT
Accuracies(%)	C1	**92.84±4.00**	88.34±4.26	85.44±4.14	90.98±3.36	89.74±5.90	90.42±3.06	86.82±4.05	88.11±4.90	92.28±3.55
	C2	85.06±3.75	75.32±14.01	74.63±6.16	85.46±2.77	**88.71±3.47**	76.52±4.33	82.84±2.51	84.35±2.26	86.12±3.35
	C3	**95.36±2.87**	90.43±3.63	91.38±4.49	91.55±1.95	93.44±3.59	88.22±3.11	89.40±2.04	94.21±2.23	93.64±2.94
	C4	**99.23±0.83**	98.15±1.39	97.69±1.21	98.71±0.80	97.12±1.48	97.74±1.45	97.68±0.86	98.42±0.90	97.86±1.32
	C5	99.32±0.72	99.53±1.10	96.74±2.62	98.60±1.41	**99.78±0.20**	99.22±0.80	95.12±3.11	99.61±0.69	99.34±1.10
	C6	86.89±2.58	75.46±9.11	71.94±6.49	81.18±4.26	82.46±4.21	71.63±5.54	75.14±3.68	81.48±5.69	**88.55±3.20**
	C7	**95.62±2.99**	93.57±4.03	92.67±2.86	96.04±2.16	91.84±4.76	94.88±2.24	92.51±2.11	94.62±3.20	94.91±1.90
	C8	**82.89±4.65**	77.60±4.69	69.62±3.31	77.00±5.31	79.14±5.31	72.62±2.54	73.59±3.49	81.72±5.36	79.80±3.76
	C9	**85.59±4.21**	82.52±4.57	68.38±8.47	84.21±5.02	76.91±7.53	77.60±5.86	77.37±6.04	84.25±3.01	85.54±4.24
	C10	**98.68±0.55**	95.97±4.38	94.72±1.49	95.86±1.65	93.53±5.11	92.94±4.13	92.41±3.74	97.26±1.23	97.66±2.04
	C11	**96.79±2.62**	89.37±3.94	87.56±4.83	93.28±2.49	88.45±2.44	85.16±9.88	85.75±5.08	94.29±2.86	92.42±2.07
	C12	**97.37±1.82**	86.43±9.80	75.25±7.97	94.49±4.91	96.91±2.08	87.74±5.87	78.34±6.64	97.21±2.43	96.50±2.97
	C13	72.47±4.69	65.26±12.73	58.12±4.75	70.24±4.05	70.71±7.06	61.61±5.66	59.95±5.25	73.05±6.78	**75.16±6.92**
	C14	86.41±4.89	80.35±8.87	73.63±5.48	82.35±5.15	85.22±3.60	78.46±4.05	71.88±6.67	87.40±2.23	**86.93±3.87**
	C15	**96.80±2.14**	91.40±6.52	87.71±3.57	95.53±2.64	96.09±2.61	91.92±3.06	87.79±5.87	92.84±3.73	95.60±3.60
	C16	**98.32±0.79**	94.41±3.32	97.13±1.15	98.02±0.81	96.21±2.20	96.00±1.53	94.57±4.55	96.41±3.45	98.27±0.62
OA(%)		**92.41±0.65**	87.21±2.17	85.22±1.47	90.56±1.04	89.80±1.15	86.84±1.00	85.77±1.45	90.51±1.54	91.88±0.53
AA(%)		**91.85±0.42**	86.51±1.96	82.66±1.60	89.59±0.84	89.14±0.76	85.17±1.05	83.82±0.95	90.33±0.96	91.29±0.42
κ		**0.9116±0.007**	0.8516±0.025	0.8282±0.017	0.8900±0.012	0.8814±0.013	0.8469±0.011	0.8350±0.016	0.8897±0.018	0.9054±0.006

with the poorest performance. Furthermore, both SSFTT and MASSFormer are lightweight models, and their performance is highly similar on the HanChuan dataset.

Finally, let us make a comparison of the visual effects of the classification maps obtained by various models. Similarly, we set up a random seed (42) to ensure that each model utilized the same training sample and obtained the corresponding classification map. The obtained classification maps are exhibited in Fig. 3. From a visual perspective, the classification map obtained by A2S2K-RN remains the clearest and possesses the best texture details. Additionally, the classification map obtained by MViT also possesses excellent visual effects, which further verifies its outstanding generalization performance. In contrast, the classification maps generated by other methods contain a substantial amount of noise points.

D. Experiment 3: HongHu Dataset

The scene of the HongHu dataset was acquired in 2017, it is a complex agricultural zone and contains 22 land-cover types. The size of the imagery is 940 × 475, there are 270 bands from 400 to 1000 nm, and the spatial resolution is about 0.043 m. Fig. 4(a) and (b) gives the pseudo-color composite image and the ground truth.

Accordingly, the classification results obtained by various models on the HongHu dataset are presented in Table III. From the data given in Table III, we can see that the proposed MViT achieved the best classification performance on the HongHu dataset, and it is the sole model with an average OA of over 93%. In operational terms, MViT takes a rather less obvious feature of ViT (or one of ViT's strengths): MViT is capable of processing sequence data with varying sequential lengths. Precisely due to the introduction of the random masking operations during the training process, MViT's training speed becomes incredibly fast. The HongHu dataset is an extremely challenging one, containing numerous types of ground objects and having a very high proportion of labeled pixels. When we employ a model to undertake a classification task on such a dataset, the performance of the model is of significance, but its complexity is equally crucial. MViT can not only achieve highly satisfactory classification results, but also has a very low

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:20:51 UTC from IEEE Xplore. Restrictions apply.

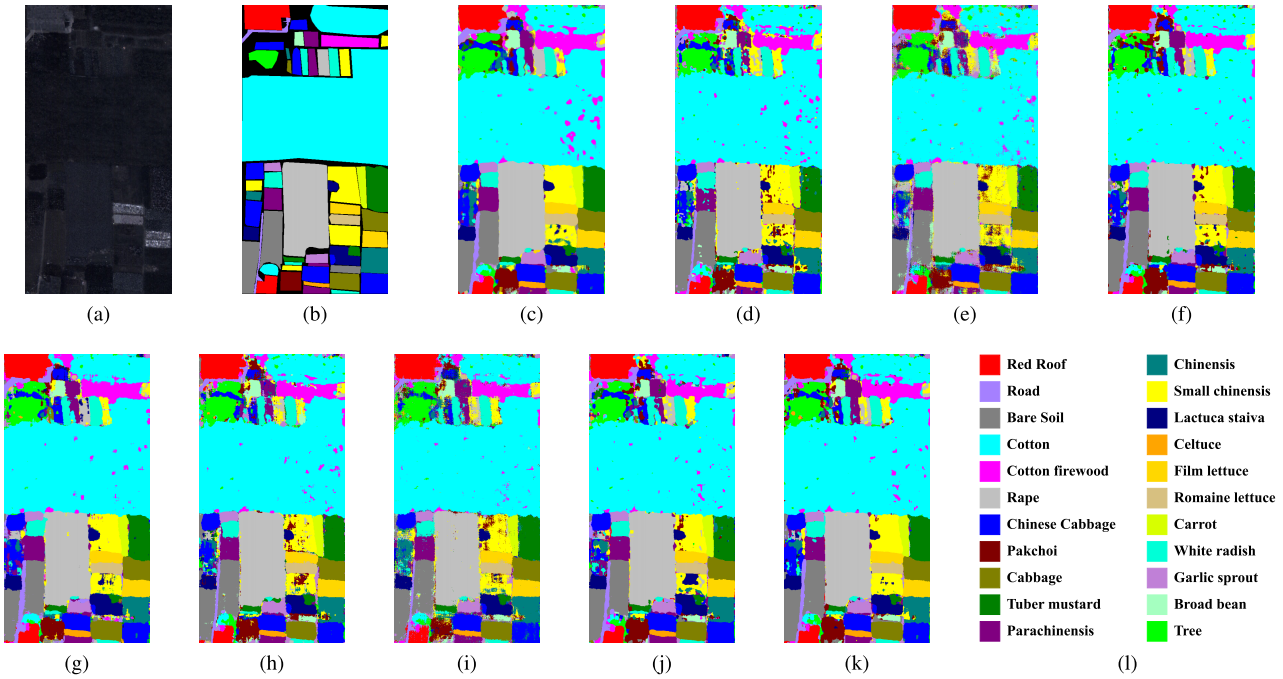


Fig. 4. Visualization of the classification maps obtained by different models on the HongHu dataset. (a) Three-band false color composite. (b) Ground truth. (c)–(k) Classification maps obtained by A2S2K-RN, CapsViT, SpectralFormer, SSFTT, GAHT, HybridFormer, SQSFormer, MASSFormer, and MViT. (l) Legend.

TABLE III
CLASSIFICATION RESULTS OF VARIOUS MODELS ON THE HONGHU DATASET

Class		A2S2K-RN	CapsViT	SpectralFormer	SSFTT	GAHT	HybridFormer	SQSFormer	MASSFormer	MViT
Accuracies(%)	C1	95.19±2.53	94.54±3.88	91.09±2.93	92.31±3.34	94.99±2.20	94.70±1.83	93.97±1.96	93.81±2.32	**95.25±1.52**
	C2	95.62±2.36	94.46±1.71	86.32±4.90	94.73±3.83	89.11±6.12	91.53±3.29	88.00±3.78	93.95±3.73	**96.32±2.03**
	C3	**88.25±2.64**	87.95±2.46	81.70±3.60	86.36±2.91	87.18±2.63	84.52±2.15	82.04±4.67	84.94±4.14	88.10±3.03
	C4	95.93±2.52	95.90±1.51	95.69±0.73	96.74±0.87	**97.14±1.21**	95.86±1.65	95.85±1.24	95.70±1.98	96.89±1.08
	C5	**97.39±2.35**	94.86±2.03	94.92±1.44	93.69±3.34	97.30±1.56	95.66±0.88	96.19±1.63	96.61±1.92	95.40±1.52
	C6	94.91±1.84	93.88±2.23	93.04±2.41	93.65±1.97	**95.54±1.31**	93.64±2.33	92.26±2.08	94.53±1.58	93.78±2.39
	C7	83.12±5.51	81.94±4.44	68.98±6.71	79.99±6.38	79.54±2.41	73.93±4.79	71.14±5.42	82.29±4.77	**85.15±1.93**
	C8	89.90±5.87	82.23±5.92	75.15±7.22	84.23±4.92	91.72±3.58	81.62±6.27	78.31±8.79	90.57±4.19	**92.91±3.44**
	C9	**97.53±1.71**	96.86±1.85	95.01±1.48	96.73±1.72	95.62±2.61	95.38±2.10	95.93±1.28	96.66±1.71	96.47±2.08
	C10	90.24±2.27	89.03±2.02	81.57±2.07	87.87±2.70	91.46±3.15	87.27±4.00	86.71±5.15	90.79±4.04	**93.58±2.18**
	C11	89.75±5.66	83.74±5.44	73.97±7.14	88.45±4.79	88.52±4.51	78.25±4.11	83.61±3.77	89.28±3.79	**91.97±3.99**
	C12	**85.14±5.10**	84.06±4.09	71.44±3.20	78.08±3.79	82.36±4.57	73.48±4.23	80.00±4.98	81.26±6.00	84.66±4.49
	C13	76.11±3.44	73.37±4.37	66.18±4.56	74.50±3.35	76.79±4.12	64.94±7.08	71.51±4.93	77.42±3.75	**79.72±4.16**
	C14	**95.95±1.49**	93.48±2.56	87.90±4.63	92.78±4.93	93.72±5.21	88.73±4.35	94.19±3.67	95.72±1.13	95.60±2.86
	C15	99.42±0.77	99.28±0.56	97.07±1.77	**99.88±0.21**	99.58±0.49	98.97±1.36	99.01±0.94	99.37±0.70	99.42±0.94
	C16	96.48±2.40	97.24±1.75	92.54±2.76	95.04±2.89	**97.81±1.73**	95.76±2.04	95.68±1.84	97.48±1.66	97.46±1.48
	C17	**99.26±0.89**	97.90±1.44	93.65±3.99	96.72±6.49	99.06±0.72	97.55±1.11	96.86±3.48	98.40±1.60	99.04±0.91
	C18	**99.25±0.38**	98.62±1.15	97.52±0.92	97.74±1.35	96.88±2.16	98.36±0.77	96.80±1.64	98.10±1.00	97.41±1.97
	C19	**96.73±1.72**	93.56±2.01	90.30±3.00	92.64±2.44	89.97±4.57	91.38±2.91	91.98±2.49	94.14±1.95	92.72±3.37
	C20	98.29±0.97	97.41±1.17	94.83±1.13	96.93±2.01	96.10±1.46	97.84±2.42	95.06±2.78	97.30±1.69	**98.48±1.13**
	C21	**99.87±0.39**	99.62±0.76	97.47±2.01	98.59±1.80	96.71±2.69	98.87±1.36	95.99±2.00	98.20±2.10	98.95±1.14
	C22	**98.95±1.91**	97.67±3.63	97.86±1.05	96.27±3.07	97.46±2.54	96.92±3.26	96.69±2.30	98.45±1.97	98.85±1.58
OA(%)		92.95±1.10	92.02±0.58	88.78±0.47	91.96±0.43	92.89±0.67	90.10±0.87	90.28±0.87	92.39±0.91	**93.65±0.53**
AA(%)		93.79±0.42	92.16±0.74	87.46±0.55	91.54±0.56	92.48±0.67	89.78±0.66	89.90±0.94	92.95±0.50	**94.05±0.36**
κ		0.9118±0.013	0.9001±0.007	0.8596±0.006	0.8992±0.005	0.9108±0.008	0.8761±0.011	0.8784±0.011	0.9048±0.011	**0.9203±0.006**

model complexity. That is why we strongly recommend it. The second-best performance on the HongHu dataset was achieved by A2S2K-RN. From the perspective of OA, A2S2K-RN is only 0.7% lower than MViT. Therefore, we highly commend the capacity of A2S2K-RN to acquire distinguishable features. Moreover, it is evident that the performance of GAHT is comparable to that of A2S2K-RN, with GAHT achieving the third-best results. CapsViT and SSFTT demonstrated comparable performance, both attaining approximately 92% OA. Of course, MASSFormer has indeed proven capable of obtaining satisfactory classification results on the HongHu dataset. HybridFormer and SQSFormer demonstrate limited performance, achieving an average OA of slightly over 90%, which suggests that they lack sufficient robustness.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:20:51 UTC from IEEE Xplore. Restrictions apply.

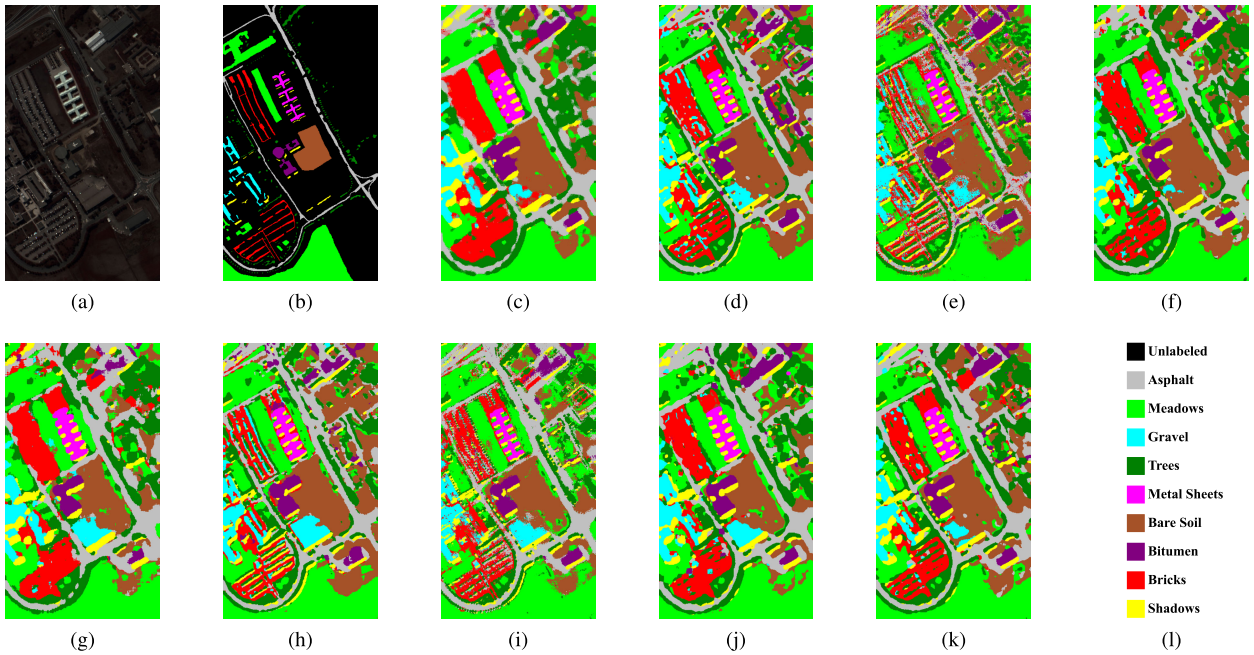


Fig. 5. Visualization of the classification maps obtained by different models on the PU dataset. (a) Three-band false color composite. (b) Ground truth. (c)–(k) Classification maps obtained by A2S2K-RN, CapsViT, SpectralFormer, SSFTT, GAHT, HybridFormer, SQSFormer, MASSFormer, and MViT. (l) Legend.

TABLE IV
CLASSIFICATION RESULTS OF VARIOUS MODELS ON THE PU DATASET

Class		A2S2K-RN	CapsViT	SpectralFormer	SSFTT	GAHT	HybridFormer	SQSFormer	MASSFormer	MViT
Accuracies(%)	C1	91.40±2.35	93.63±3.01	78.04±6.15	90.16±3.95	90.04±2.14	92.15±3.17	91.99±3.00	94.20±3.81	**95.97±3.29**
	C2	**97.95±1.28**	96.99±1.54	94.62±2.80	97.43±1.47	96.64±1.22	94.06±2.80	96.71±3.02	96.07±2.77	97.73±1.55
	C3	**94.12±2.94**	91.78±4.74	86.18±3.91	93.42±4.65	92.13±4.14	89.24±3.77	88.14±2.55	93.15±3.33	93.70±2.58
	C4	90.64±3.87	96.61±1.53	94.52±2.08	92.67±3.03	90.54±3.38	95.04±1.42	**97.62±1.15**	93.39±2.15	96.80±1.59
	C5	98.64±1.46	**100.00±0.00**	99.74±0.51	99.42±0.33	98.68±0.72	99.92±0.09	**100.00±0.00**	99.14±0.84	99.63±0.89
	C6	**99.81±0.36**	97.34±2.26	95.10±2.50	98.83±1.02	99.26±0.74	97.97±1.79	96.90±1.39	98.94±1.15	98.60±1.70
	C7	99.16±0.73	**99.86±0.19**	99.20±1.06	99.55±0.49	98.97±1.49	99.75±0.35	99.83±0.27	99.49±1.24	99.73±0.27
	C8	92.45±4.28	90.10±7.31	66.54±8.84	91.39±1.82	**93.79±5.48**	85.76±3.57	84.65±3.80	90.81±4.05	92.48±4.76
	C9	91.44±3.78	98.85±1.07	97.32±1.74	97.54±1.20	91.52±3.03	96.19±4.78	**99.92±0.13**	97.97±1.13	99.01±0.68
OA(%)		95.89±0.74	95.86±1.14	89.63±1.93	95.54±1.15	95.05±0.89	93.72±0.76	94.95±1.50	95.57±1.12	**96.99±0.96**
AA(%)		95.07±0.76	96.13±1.30	90.14±1.09	95.60±0.98	94.62±0.86	94.44±0.89	95.14±0.68	95.91±0.80	**97.07±0.82**
κ		0.9457±0.010	0.9453±0.015	0.8639±0.024	0.9411±0.015	0.9348±0.012	0.9176±0.010	0.9334±0.019	0.9418±0.014	**0.9603±0.013**

Finally, based on the classification results, we present the classification graphs obtained by each model on the HongHu dataset (all models used the same training samples). The obtained classification maps are shown in Fig. 4. It is clearly observable from Fig. 4 that MViT has a superior classification effect for category 13 (Small chinensis, light yellow), while A2S2K-RN demonstrates a better classification effect for category 6 (Rape, light gray). In terms of the overall classification effect, both MViT and A2S2K-RN are superior to other approaches. Furthermore, it can be observed that the classification map obtained by GAHT also has an excellent visual effect, which validates that it can achieve satisfactory classification results on the HongHu dataset.

E. Experiment 4: PU Dataset

The PU dataset is one of the most classic classification datasets within the remote sensing field. This dataset was obtained by the reflective optics system imaging spectrometer (ROSIS) sensor during a flight activity over the PU in 2001. Twelve bands covering noise and water absorption have been discarded, thus only 103 bands of data are retained, and the wavelength ranges from 430 to 860 nm. This dataset depicts the ground scene in a 610 × 340 image format, featuring a spatial resolution of 1.3 m, and encompasses 9 types of urban land covers. Fig. 5(a) and (b) presents the three-band false color image and the ground truth.

We give the classification results in Table IV. It can be seen that MViT achieved the optimal classification performance on the PU dataset, and demonstrated relatively low standard deviations across the three evaluation metrics (0.96%, 0.82%, and 0.013, respectively), indicating that it has good stability. Due to its inherent nature, MViT acquires patterned features (the underlying patterns), thus being less influenced by the selection of training samples and performing more stably. Furthermore, from the perspective of Dropout, MViT effectively avoids overfitting problems to a certain extent, which is also one of its remarkable qualities. It is notable that A2S2K-RN and CapsViT achieved similar values of OA

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:20:51 UTC from IEEE Xplore. Restrictions apply.

TABLE V
COMPARISON OF THE COMPUTATIONAL COSTS ASSOCIATED WITH VARIOUS MODELS ON THE PU DATASET

Models	Trainable Parameters	FLOPs(10^6)	GPU Memory(MB)	Training Time(s)	Testing Time(s)	Training w/ Validation(s)
A2S2K-RN	109668	72.245	1027	27.31	2.83	23.22
CapsViT	1321416	33.817	5771	114.62	16.58	73.73
SpectralFormer	152994	11.436	443	37.03	2.94	30.91
SSFTT	152769	16.907	457	15.84	1.50	11.44
GAHT	635081	143.194	1443	28.43	4.90	23.44
HybridFormer	2060682	41.764	491	79.60	6.17	53.19
SQSFormer	171996	37.488	1017	28.56	3.23	25.33
MASSFormer	245673	29.812	601	19.19	1.83	15.80
MViT	280185	34.994	521	22.72	2.02	16.75

and κ, but CapsViT attained significantly higher AA than A2S2K-RN. The performance of SSFTT closely resembles that of MASSFormer. Likewise, the performance of GAHT is quite comparable to that of SQSFormer. As we can see that SQSFormer merely excels on the LongKou dataset, but its performance on the PU dataset is lackluster. In contrast, MViT can consistently achieve outstanding or even the best performance on various datasets. We say that a robust classification model should remain unaffected by the quantity of training samples and the variations in datasets, while maintaining a complexity that is as minimal as possible. Evidently, MViT meets the conditions.

Accordingly, the classification graphs obtained by various models on the PU dataset are presented in Fig. 5. As can be seen, the classification map obtained by MViT has few noise points and smooth textures. Particularly for categories 8 (Bricks, red) and 9 (Shadows, yellow), the visual effects are significantly better. In contrast, the classification map obtained by A2S2K-RN is overly smooth, making it challenging to distinguish the junctions of different land covers. The classification graphs obtained by SpectralFormer and SQSFormer are replete with a considerable amount of salt and pepper noise, whereas the classification maps obtained by CapsViT, SSFTT, and MASSFormer have more favorable visual effects. Overall, MViT not only achieves the best classification results but also presents the best visuals.

IV. DISCUSSION

To gain a deeper understanding of the effectiveness and superiority of MViT, we will present additional experimental results to discuss the complexity and sensitivity of MViT. All the analysis experiments were conducted on the PU dataset since this dataset is one of the most classic datasets in the remote sensing community and has an appropriate proportion of labeled pixels.

A. Complexity

For various datasets, the sample format processed by each model is fixed. Hence, it is essential to introduce the sample composition of the PU dataset in detail before conducting the complexity analysis. For the PU dataset, it comprises 225 training samples (with 25 samples randomly selected for each of the 9 classes), 900 validation samples (with 100 samples randomly selected for each class), and 42 551 test samples. Notably, the validation samples are derived from the test samples through a process of random sampling with replacement.

The complexity metrics analyzed encompass the number of trainable parameters, FLOPs, GPU memory usage, training time, and testing time. Additionally, given that each model conducts a validation on the validation set following each training epoch to retain the optimal model parameters, we also provide the training time for each model when it is simply trained for 100 epochs without performing validation (denoted as Training w/ Validation). All statistical times are measured in seconds (s), and all statistical data are presented in Table V.

As illustrated by the data presented in Table V, A2S2K-RN possesses the fewest trainable parameters (109 668) among the models under consideration. MViT has a total of 280 185 trainable parameters, when compared to the lightweight model, MASSFormer, it is evident that MViT's trainable parameter count exceeds that of MASSFormer by only 34 512. At the same time, MViT exhibits relatively low FLOPs and occupies little GPU memory. Training time serves as a critical metric for assessing the lightweight nature of a model. HybridFormer possesses the largest number of trainable parameters, yet its FLOPs and GPU occupancy are at a moderate level, and of course, its training and testing times are relatively extended due to the incorporation of multigranularity branches. Lightweight SSFTT requires merely 15.84 s for training. In comparison with the lightweight MASSFormer, MViT's training duration extended by only 3.53 s, while its testing phase only took an additional 1.9 s. It is noteworthy that when MASSFormer is applied to the PU dataset, it adjusts the number of ViT encoders from the default of 3 to 2. When MASSFormer utilizes 3 ViT encoders, it requires a longer training and testing time compared to MViT. These figures sufficiently underscore the lightweight characteristics of MViT.

We all appreciate lightweight models, and MViT is not only sufficiently lightweight but also exhibits remarkably high performance.

B. Impact of the Masking Ratio

For MViT, the masking operations play a crucial role in its ability to learn patterned features. MViT employs a default masking ratio of 75%, which is a hyperparameter that can be adjusted as needed. Here, we examine masking ratios ranging from 0% to 90% to investigate the effects of varying masking

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:20:51 UTC from IEEE Xplore. Restrictions apply.

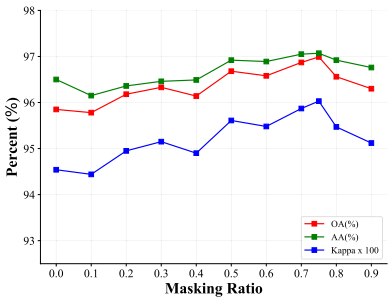


Fig. 6. Effects of varying masking ratios on MViT on the PU dataset.

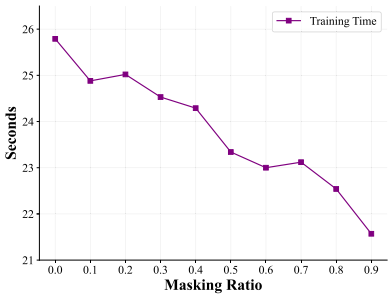


Fig. 7. Training time of MViT on the PU dataset under varying masking ratios.

ratios. For various masking ratios, we systematically selected training samples at random and conducted ten experiments to report the AA. The experimental results are shown in Fig. 6. Meanwhile, to gain a more intuitive understanding of the impact of different masking ratios on the training speed of MViT, we illustrate the training time of MViT under varying masking ratios in Fig. 7.

From the experimental results presented in Fig. 6, it is evident that, in addition to the inherent influence of the dataset on the classification performance, varying masking ratios exert distinct impacts on the performance of MViT. When the masking ratio is set to 0%, MViT transforms into a conventional model focused on learning discriminative features. In this scenario, the performance of MViT is suboptimal, suggesting that merely enabling MViT to learn conventional discriminative features does not result in superior performance. As the masking ratio increases, the performance of MViT progressively improves. When the masking ratio reaches 70%, the OA, AA, and κ values achieved by MViT are 96.87%, 97.05%, and 0.9587, respectively. At a masking ratio of 75%, MViT achieved peak performance with OA, AA, and κ values of 96.99%, 97.07%, and 0.9603, respectively. As the masking ratio continues to increase, the performance of MViT exhibits a declining trend. The aforementioned phenomena indicate that at low masking ratios, MViT fails to learn effective patterned features, while at excessively high masking ratios, the patterned features acquired by MViT lack precision. Consequently, we advise against altering the default 75% masking ratio in practical applications, as this setting typically yields optimal performance for MViT across various datasets. Finally, as shown in Fig. 7, with the gradual increase in the masking ratio, the training time required by MViT progressively decreases. This can primarily be attributed to

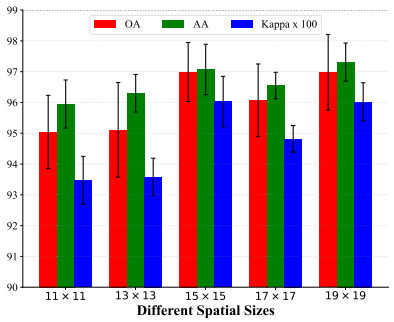


Fig. 8. Effects of varying spatial sizes of samples on MViT on the PU dataset.

the fact that an increase in the masking ratio leads to a reduction in the number of tokens processed by the 3 ViT blocks. Undoubtedly, a 75% masking ratio guarantees that MViT upholds an exceptionally efficient training.

C. Impact of the Spatial Size of Samples

As mentioned in Section III-A, an excessively large spatial size of samples may lead to a substantial increase in computational requirements for overlapping sample areas, while an overly small spatial size may limit the spatial information encapsulated within the samples. Consequently, MViT adopts a default spatial size of 15 × 15 as a balanced choice. Here, we further evaluate the impact of varying spatial sizes of samples on the performance of MViT. The spatial sizes under comparison encompass grids of 11 × 11, 13 × 13, 15 × 15, 17 × 17, and 19 × 19, and Fig. 8 presents the experimental results.

As depicted in Fig. 8, when the spatial size of the training sample is limited, the classification accuracy of MViT tends to be relatively low. This is attributed to the restricted spatial information within the sample due to its small spatial size, which prevents the model from effectively learning discriminative patterned features. In addition, it is not always true that a larger spatial size leads to better performance of MViT. Specifically, when the spatial size of the sample is set to 17 × 17, the performance of MViT is inferior to that achieved with a spatial size of 15 × 15. We hypothesize that this phenomenon may be attributed to the inherent characteristics of the PU dataset. In other words, dividing the samples using a 17 × 17 grid on the PU dataset does not yield the optimal performance for MViT. Finally, when the spatial size is set to 19 × 19, the performance of MViT becomes nearly equivalent to that achieved with a spatial size of 15 × 15, and in some cases, even surpasses the AA value obtained at the 15 × 15 setting. However, it is crucial to highlight that an increase in the spatial size of samples will increase the FLOPs of the model. Additionally, for datasets with a high proportion of labeled pixels, such as the WHU-Hi dataset, a larger spatial size of samples will also result in higher memory consumption.

To summarize, in practical applications, we recommend utilizing samples with a spatial size of 15 × 15 as the input for the model. This is because such a spatial size typically enables the model to achieve good classification performance while maintaining relatively low computational costs.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:20:51 UTC from IEEE Xplore. Restrictions apply.

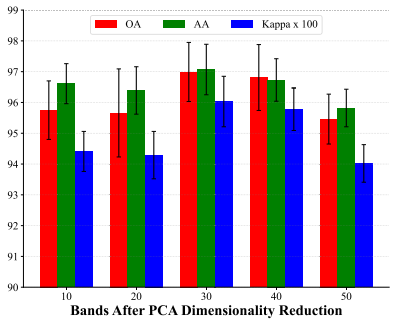


Fig. 9. Effects of the PCA dimensionality reduction on MViT on the PU dataset.

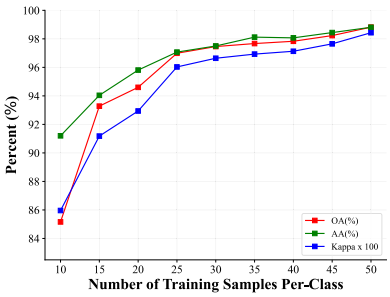
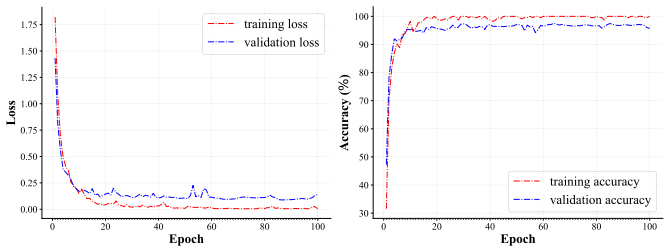
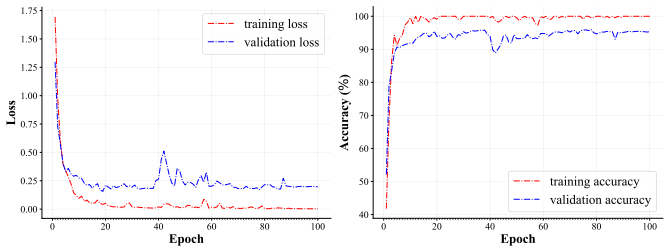


Fig. 10. Effects of the number of training samples on MViT on the PU dataset.



(a) (b)

Fig. 11. Loss and accuracy curves of MViT on both the training and validation sets on the PU dataset. (a) Loss curves. (b) Accuracies curves.



(a) (b)

Fig. 12. Loss and accuracy curves of MViT when the central embedding is randomly masked. (a) Loss curves. (b) Accuracies curves.

D. Impact of the PCA Dimensionality Reduction

Any dimensionality reduction technique inevitably results in some degree of information loss. To ensure the training and testing efficiency of the model, MViT employs PCA as the default dimensionality reduction method to reduce the original HSI data from its full dimensionality to 30 bands. In this section, we will further examine the impact of PCA dimensionality reduction on the performance of MViT. We set the number of bands retained after PCA dimensionality reduction to 10, 20, 30, 40, and 50, respectively, to evaluate the performance of MViT, and the experimental results are presented in Fig. 9.

As depicted in Fig. 9, when the number of retained bands is set to 10 or 20, the classification accuracy achieved by MViT remains relatively low. This can be attributed to the significant loss of spectral information resulting from PCA dimensionality reduction. When the number of retained bands is increased to 30 or 40, MViT achieved highly accurate classification results. However, when the number of reserved bands is set to 50, the performance of MViT is even worse than that when the number of reserved bands is set to 10 or 20. This is primarily due to the fact that the 2-D and 3-D convolutions employed by MViT utilize a limited number of convolutional kernels, which are insufficient for effectively extracting spectral–spatial features, thereby causing the model to struggle in learning separable patterned features. It is imperative to emphasize that an excessive number of reserved bands can not only lead to the curse of dimensionality but also consume a significant amount of storage space.

In conclusion, retaining 30 bands not only allows MViT to complete the training and testing processes more efficiently but also enables it to achieve outstanding classification performance.

E. Impact of the Number of Training Samples

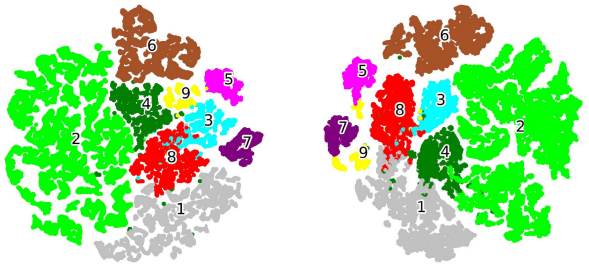
We incrementally increased the number of samples per class from 10 to 50 to investigate the impact of varying training sample sizes on the performance of MViT. The experimental results are shown in Fig. 10.

As illustrated by the experimental results presented in Fig. 10, the performance of MViT consistently enhances as the sample size for each category grows, suggesting that the patterned features learned by MViT become increasingly precise when the number of training samples increases. As a conclusion, similar to the existing classification models that emphasize the learning of the discriminative features, MViT demonstrates consistent improvement as the number of training samples progressively increases.

F. Stability

During the training phase, although MViT employs a random masking operation with a 75% masking ratio to obscure the majority of embedding features, there is no need for concern, MViT maintains a stable training process, primarily because it intentionally preserves the central embedding when performing the random masking operation. Here, we present the loss and accuracy curves of MViT on both the training and validation sets to assess the stability of its training process. Please refer to the loss and accuracy curves depicted in Fig. 11. In addition, for comparison and to further emphasize the significance of retaining the central embedding, we also give the loss and accuracy curves of MViT when the central embedding is randomly masked, as shown in Fig. 12. (In the above two scenarios, MViT used the same training samples for training).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:20:51 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 13. Patterned features learned by MViT and the discriminative features extracted by A2S2K-RN from the test samples of the PU dataset. (a) Patterned features learned by MViT. (b) Discriminative features extracted by A2S2K-RN.

First, as illustrated in Fig. 11, it is clear that MViT can be trained with stability akin to that of a conventional classifier. Second, as depicted in Fig. 12, it is evident that when the central embedding is randomly discarded, although MViT remains capable of converging, its training process becomes oscillatory, and the accuracy achieved on the validation set also diminishes. In Section II-B, we elucidate from a design standpoint why the central embedding cannot be masked. In this context, we reiterate the critical importance of deliberately maintaining the center embedding throughout the training process. This strategy not only underscores the significance of the central pixel in HIC but also guarantees the model's efficacy and ensures its stable training.

G. t-SNE Visualization

To visually assess the separability of the patterned features acquired by MViT, we present the 2-D t-distributed stochastic neighbor embedding (t-SNE) visualization of the patterned features extracted by MViT from the test samples of the PU dataset in Fig. 13(a).[2] Considering that A2S2K-RN achieved the second-highest classification performance on the PU dataset, for the purpose of comparison, we also provide the 2-D t-SNE visualization of the discriminative features obtained by A2S2K-RN from the test samples in Fig. 13(b)[3] (MViT and A2S2K-RN were trained using the same training samples).

It can be observed from Fig. 13 that among the discriminative features learned by A2S2K-RN, a relatively substantial aliasing effect exists between the first and third classes, whereas the patterned features learned by MViT do not exhibit this phenomenon. Furthermore, for the samples belonging to the ninth category, the patterned features extracted by MViT exhibit markedly greater cohesiveness in comparison to the discriminative features learned by A2S2K-RN. Certainly, for the samples belonging to the second and fourth classes, the patterned features extracted by MViT demonstrate superior separability in comparison to the discriminative features learned by A2S2K-RN.

[2]The patterned features learned by MViT correspond to the class tokens output by the third ViT block.

[3]Here, the 2-D t-SNE visualizations of the discriminative features learned by other classification models will not be presented. Readers who are interested may conduct their own comparisons.

In summary, unlike the existing classification models that depend on the learning of the discriminative features, MViT learns more robust patterned features, and these patterned features typically exhibit superior separability compared to the discriminative features. This is the underlying reason why MViT is capable of achieving either competitive or state-of-the-art classification performance across various datasets. As the pioneering classifier based on patterned feature learning, we anticipate that MViT will make significant contributions to the remote sensing community.

V. Conclusion

In this article, we put forward a lightweight MViT for HIC. Through introducing the masking operations in the training phase, MViT places emphasis on the learning of patterned features instead of discriminative features, thereby endowing it with superior generalization performance. Furthermore, to emphasize the contribution of the central pixel in HIC and ensure the effectiveness of the entire model, MViT purposefully retains the embedded feature associated with the central pixel when conducting the masking operations in the training phase. Importantly, MViT possesses a straightforward structure, and due to the introduction of the masking operations, its training speed is also extremely fast. During the testing phase, MViT will deactivate the masking operations and employ all the embedded features to fulfill the classification task, which endows it with more potent recognition capabilities. Extensive experiments have demonstrated that, in contrast to the state-of-the-art approaches, MViT consistently attains outstanding or even optimal classification results. Given that the currently developed MViT primarily emphasizes the learning of patterned features within the spatial domain, future efforts may explore spectral–spatial dual-masking strategies to acquire more robust spectral–spatial joint patterned features. Furthermore, in light of A2S2K-RN and HybridFormer, it is evident that multiscale patterned feature learning constitutes a domain deserving further investigation.

References

[1] Q. Sun, G. Zhao, Y. Fang, C. Fang, L. Sun, and X. Li, "MEA-EFFormer: Multiscale efficient attention with enhanced feature transformer for hyperspectral image classification," *Remote Sens.*, vol. 16, no. 9, p. 1560, Apr. 2024.

[2] M. Q. Alkhatib et al., "Tri-CNN: A three branch model for hyperspectral image classification," *Remote Sens.*, vol. 15, no. 2, p. 316, Jan. 2023.

[3] M. Wang, Y. Sun, J. Xiang, R. Sun, and Y. Zhong, "Adaptive learnable spectral–spatial fusion transformer for hyperspectral image classification," *Remote Sens.*, vol. 16, no. 11, p. 1912, May 2024.

[4] Y. Li et al., "MEET: A million-scale dataset for fine-grained geospatial scene classification with zoom-free remote sensing imagery," 2025, *arXiv:2503.11219*.

[5] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3D-2D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.

[6] S. Ding, X. Ruan, J. Yang, J. Sun, S. Li, and J. Hu, "LSSMA: Lightweight spectral–spatial neural architecture with multiattention feature extraction for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 6394–6413, 2024.

[7] J. Ma, Y. Zou, X. Tang, X. Zhang, F. Liu, and L. Jiao, "Spatial pooling transformer network and noise-tolerant learning for noisy hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5509719.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:20:51 UTC from IEEE Xplore. Restrictions apply.

[8] G. Tang, X. Wang, H. Zhao, X. Hu, G. Jin, and Y. Zhong, "Attention in attention for hyperspectral with high spatial resolution (H) image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5501113.
[9] Q. Zhu et al., "A spectral–spatial-dependent global learning framework for insufficient and imbalanced hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 52, no. 11, pp. 11709–11723, Nov. 2022.
[10] L. Wang, H. Wang, L. Wang, X. Wang, Y. Shi, and Y. Cui, "RSSGL: Statistical loss regularized 3-D ConvLSTM for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5529420.
[11] C. Fu, B. Du, and L. Zhang, "Do we need learnable classifiers? A hyperspectral image classification algorithm based on attention-enhanced ResBlock-in-ResBlock and ETF classifier," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5505013.
[12] S. Qiao, X.-M. Dong, J. Peng, and W. Sun, "LiteSCANet: An efficient lightweight network based on spectral and channel-wise attention for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 11655–11668, 2021.
[13] C. Wu, L. Tong, J. Zhou, and C. Xiao, "Spectral–spatial large kernel attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5508814.
[14] S. K. Roy, S. Manna, T. Song, and L. Bruzzone, "Attention-based adaptive spectral–spatial kernel ResNet for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7831–7843, Sep. 2021.
[15] Z. Meng, L. Jiao, M. Liang, and F. Zhao, "A lightweight spectral–spatial convolution module for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[16] Y. Chen et al., "A new dual-branch embedded multivariate attention network for hyperspectral remote sensing classification," *Remote Sens.*, vol. 16, no. 11, p. 2029, Jun. 2024.
[17] Z. Ye, C. Li, Q. Liu, L. Bai, and J. E. Fowler, "Computationally lightweight hyperspectral image classification using a multiscale depth-wise convolutional network with channel attention," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.
[18] X. Ma et al., "An ultralightweight hybrid CNN based on redundancy removal for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5506212.
[19] Z. Zheng, Y. Zhong, A. Ma, and L. Zhang, "FPGA: Fast patch-free global learning framework for fully end-to-end hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5612–5626, Aug. 2020.
[20] M. E. Paoletti et al., "Capsule networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2145–2160, Apr. 2019.
[21] H. Wang and L. Wang, "Collaborative active learning based on improved capsule networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5522126.
[22] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.
[23] A. Ravikumar, P. N. Rohit, M. K. Nair, and V. Bhatia, "Hyperspectral image classification using deep matrix capsules," in *Proc. Int. Conf. Data Sci., Agents Artif. Intell. (ICDSAAI)*, vol. 1, Dec. 2022, pp. 1–7.
[24] G. E. Hinton, S. Sabour, and N. Frosst, "Matrix capsules with EM routing," in *Proc. Int. Conf. Learn. Represent.*, Feb. 2018, pp. 1–15.
[25] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.
[26] S. Wan, C. Gong, P. Zhong, B. Du, L. Zhang, and J. Yang, "Multiscale dynamic graph convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3162–3177, May 2020.
[27] M. Jiang et al., "GraphGST: Graph generative structure-aware transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504016.
[28] Z. Zhao, H. Wang, and X. Yu, "Spectral–spatial graph attention network for semisupervised hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[29] A. Vaswani et al., "Attention is all you need," in *Proc. NIPS*, 2017, pp. 6000–6010.
[30] J. Gao, X. Ji, G. Chen, and R. Guo, "Main–sub transformer with spectral–spatial separable convolution for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 2747–2762, 2024.
[31] C. Shi, S. Yue, and L. Wang, "A dual-branch multiscale transformer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504520.
[32] I. Ahmad, G. Farooque, Q. Liu, F. Hadi, and L. Xiao, "MSTSENet: Multiscale spectral–spatial transformer with squeeze and excitation network for hyperspectral image classification," *Eng. Appl. Artif. Intell.*, vol. 134, Aug. 2024, Art. no. 108669.
[33] F. Xu, G. Zhang, C. Song, H. Wang, and S. Mei, "Multiscale and cross-level attention learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501615.
[34] X. He, Y. Chen, and Z. Lin, "Spatial–spectral transformer for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 3, p. 498, Jan. 2021.
[35] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.
[36] S. Zhang, J. Zhang, X. Wang, J. Wang, and Z. Wu, "ELS2T: Efficient lightweight spectral–spatial transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5518416.
[37] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.
[38] S. Mei, C. Song, M. Ma, and F. Xu, "Hyperspectral image classification using group-aware hierarchical transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5539014.
[39] E. Ouyang, B. Li, W. Hu, G. Zhang, L. Zhao, and J. Wu, "When multigranularity meets spatial–spectral attention: A hybrid transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4401118.
[40] N. Chen, L. Fang, Y. Xia, S. Xia, H. Liu, and J. Yue, "Spectral query spatial: Revisiting the role of center pixel in transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5402714.
[41] Q. Gu, H. Luan, K. Huang, and Y. Sun, "Hyperspectral image classification using multi-scale lightweight transformer," *Electronics*, vol. 13, no. 5, p. 949, Feb. 2024.
[42] L. Sun, H. Zhang, Y. Zheng, Z. Wu, Z. Ye, and H. Zhao, "MASSFormer: Memory-augmented spectral–spatial transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516415.
[43] Q. Wu, M. He, W. Huang, and F. Zhu, "Dilated deep MPFormer network for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.
[44] W. Yu et al., "MetaFormer is actually what you need for vision," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 10809–10819.
[45] J. Zhao, J. Wang, C. Ruan, Y. Dong, and L. Huang, "Dual-branch spectral–spatial attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504718.
[46] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics, Hum. Lang. Technol.*, vol. 1. Minneapolis, MI, USA: Association for Computational Linguistics, Jun. 2019, pp. 4171–4186.
[47] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 15979–15988.
[48] L. Wang, H. Wang, P. Wang, and L. Wang, "Regularized masked auto-encoder for semi-supervised hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5540521.



Liguo Wang (Member, IEEE) received the M.S. degree in natural science and the Ph.D. degree in engineering from Harbin Institute of Technology (HIT), Harbin, China, in 2002 and 2006, respectively.

From 2006 to 2008, he held a Post-Doctoral Research Position with Harbin Engineering University, Harbin. He is currently a Professor with the College of Information and Communication Engineering, Dalian Nationalities University, Dalian, China. He has published four books, 50 patents, and more than 320 articles in journals and conference proceedings. His research interests include remote sensing image processing and machine learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:20:51 UTC from IEEE Xplore. Restrictions apply.



Heng Wang received the B.S. degree in computer and communication engineering from Zhengzhou University of Light Industry, Zhengzhou, China, in 2015, and the M.S. degree in information and communication engineering from Harbin Engineering University, Harbin, China, in 2019, where he is currently pursuing the Ph.D. degree with the College of Information and Communication Engineering.

His research interests include computer vision, image classification, self-supervised representation learning, and remote sensing image processing.



Shoulin Yin received the B.S. and M.S. degrees from Shenyang Normal University, Shenyang, China, in 2013 and 2015, respectively. He is currently pursuing the Ph.D. degree with the College of Information and Communication Engineering, Harbin Engineering University, Harbin, China.

He has participated in several provincial and ministerial projects. His research interests include remote sensing image processing and object detection.

Mr. Yin also holds the positions of an editorial board member and the deputy editor of international journals, and acts as a reviewer of IEEE ACCESS and IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING.



Lifeng Wang received the B.S. and M.S. degrees from Northeast Agricultural University, Harbin, China, in 2014 and 2017, respectively, and the Ph.D. degree in information and communication engineering from Harbin Engineering University, Harbin, in 2022.

She is currently a Lecturer with the School of Technology, Beijing Forestry University, Beijing, China. She has authored or co-authored ten peer-reviewed articles in international journals, including *Remote Sensing of Environment*, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, *Remote Sensing*, and *Neural Networks*. Her research interests include hyperspectral image change detection and deep learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:20:51 UTC from IEEE Xplore. Restrictions apply.